A1. Caesar Cipher

1(a) Encrypt "SECURITY" with key = 4

Caesar Cipher rule:

$C = (P + k) \bmod 26$

Plain	Position	+4	Cipher
S	18	22	W
E	4	8	I
C	2	6	G
U	20	24	Y
R	17	21	V
I	8	12	M
T	19	23	X
Y	24	2	C

Encrypted text:
$\boxed{\text{WIGYVMXC}}$

1(b) Decrypt "WKHQ"

Decryption rule:

$P = (C - k) \bmod 26$

Cipher	−4	Plain
W	S	S
K	G	G
H	D	D
Q	M	M

Decrypted text:
$\boxed{\text{SGDM}}$

A2. Playfair Cipher

2(a) Playfair Matrix using keyword: MONARCHY

Rules:

- Combine **I/J**
- Remove duplicates

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Encrypt "INSTRUMENT"

Prepare digraphs:

`IN ST RU ME NT`

Digraph	Rule	Cipher
IN	Rectangle	GA
ST	Same row	TL
RU	Rectangle	MX
ME	Rectangle	CE
NT	Rectangle	RQ

Ciphertext:
$\boxed{\text{GATLMXCERQ}}$

2(b) Encrypt "BALLOON" (digraph formation)

`BA LX LO ON`

Digraph	Cipher
BA	DB
LX	SU
LO	PM
ON	NA

Ciphertext:
$\boxed{\text{DBSUPMNA}}$

A3. Hill Cipher

3(a) Encrypt "HELP"

Key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Plaintext values:

$H = 7,\ E = 4,\ L = 11,\ P = 15$

Vectors:

$\begin{bmatrix}7\\4\end{bmatrix},\ \begin{bmatrix}11\\15\end{bmatrix}$

First block:

$$\begin{bmatrix}3 & 3\\2 & 5\end{bmatrix}\begin{bmatrix}7\\4\end{bmatrix}$$

$$\begin{bmatrix}33\\34\end{bmatrix} \bmod 26$$

$\begin{bmatrix}7\\8\end{bmatrix} \Rightarrow \text{HI}$

Second block:

$\begin{bmatrix}6\\19\end{bmatrix} \Rightarrow \text{GT}$

Ciphertext:
$\boxed{\text{HIGT}}$

3(b) Validity of key matrix

$\det(K) = (3 \times 5 - 3 \times 2) = 9$

$\gcd(9, 26) = 1$

Valid Hill Cipher key

A4. Modular Arithmetic

4(a)

$(17 + 20) \bmod 26 = \boxed{11}$

4(b)

$(7 \times 9) \bmod 26 = \boxed{11}$

Multiplicative inverse of 5 mod 7

$5 \times 3 = 15 \equiv 1 \pmod 7$

$\boxed{5^{-1} \equiv 3 \pmod{7}}$

A5. Euclidean Algorithm

Find GCD

$26 = 7 \times 3 + 5$

$7 = 5 \times 1 + 2$

$5 = 2 \times 2 + 1$

$\boxed{\gcd(26, 7) = 1}$

Extended Euclidean Algorithm

$1 = 5 - 2 \times 2$

$= 5 - (7 - 5) \times 2$

$= 3 \times 5 - 2 \times 7$

$5 = 26 - 7 \times 3$

$1 = 3 \times 26 - 11 \times 7$

$\boxed{7^{-1} \equiv 15 \pmod{26}}$

A6. Finite Fields

Multiply in GF(2^3)

$(101) \times (011) = 1111$

Reduce using irreducible polynomial:

$x^3 + x + 1$

Final result:

$\boxed{010}$